DIY DNS DFIR: You're Doing it WRONG

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About Me

- Andrew Hay
  - Chief Information Security Officer (CISO) @ DataGravity
  - Former:
    - Director of Research @ OpenDNS
    - Chief Evangelist & Director of Research @ CloudPassage
    - Senior Security Analyst @ 451 Research
  - Wrote some books, blog occasionally, and give Rob a hard time on Twitter

Overview

- Whirlwind tour of DNS
- Why DNS is so valuable for TH and IR
- Real world example
- Available tools
- Summary

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Whirlwind tour of DNS

Hang on tight!

Overview

- The Domain Name System (DNS) is one of the most essential parts of the Internet's infrastructure
  - Some might say it's the plumbing ☺
- By using DNS, you can connect to a website without having to know the website's IP address.
  - After all, who wants to remember every website's IP address
- Recursive vs. Authoritative
  - Authoritative DNS nameservers are responsible providing answers to recursive DNS nameservers with the IP "mapping" of the intended website

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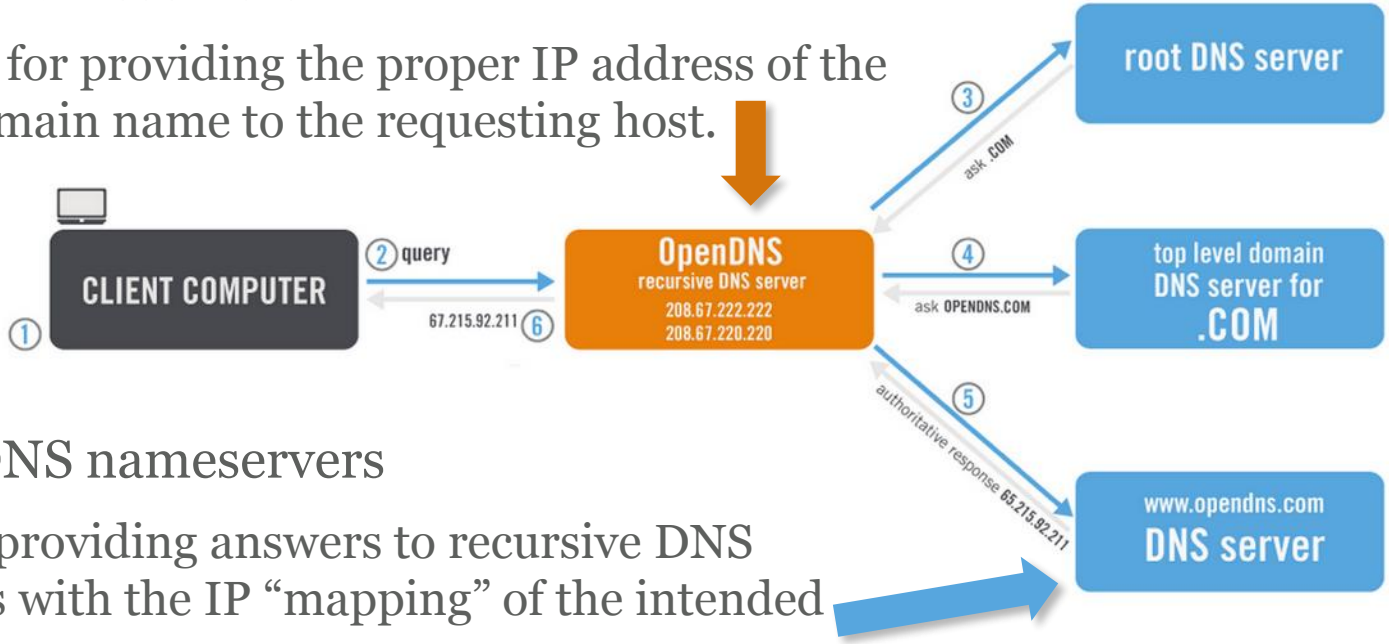
What DNS Looks Like

- Recursive DNS nameservers
  - Responsible for providing the proper IP address of the intended domain name to the requesting host.



- Authoritative DNS nameservers
  - responsible providing answers to recursive DNS nameservers with the IP “mapping” of the intended website

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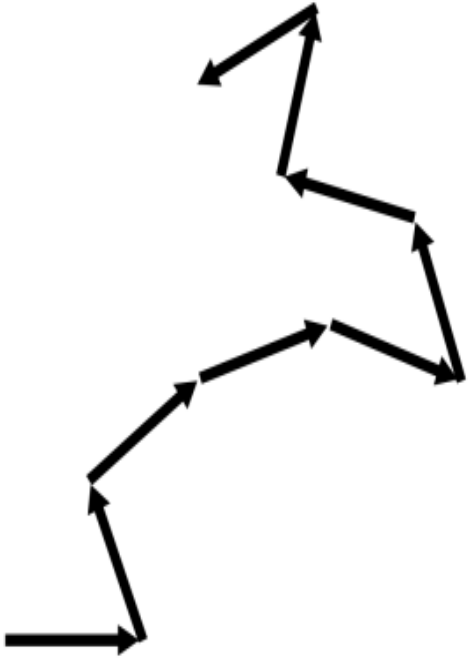
Why DNS is so valuable

For threat hunting and incident response

Why DNS is so valuable

- The more data you have, the better suited you are to **CONSTRUCT A TIMELINE** of events
- At the **VERY LEAST** you should be able to **MAKE MORE INFORMED DECISIONS**

- DNS traffic is often the most overlooked and undervalued network-based data source



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Why DNS is so valuable

- DNS shows you…
  - What domains are being looked up by humans and machines
  - The frequency at which domains are being looked up
  - What parts of the world (based on TLD) your systems are trying to hit
  - Wheather or not the domains being queried are live
  - If subsequent queries (redirects) are made as a result of a particular query
  - Wheather or not the queries are attempting to resolve fradulent/phishing sites

Why DNS is so valuable – DGA Detection

- Domain Generation Algorithm (DGA)
  - Used to generate domains programmatically
- Typically rely on a seed of some sort
  - Date, time, keyword, etc.
- Allows for the registration of domains that no human would ever type

more funny stuff at FUNNYASDUCK.NET

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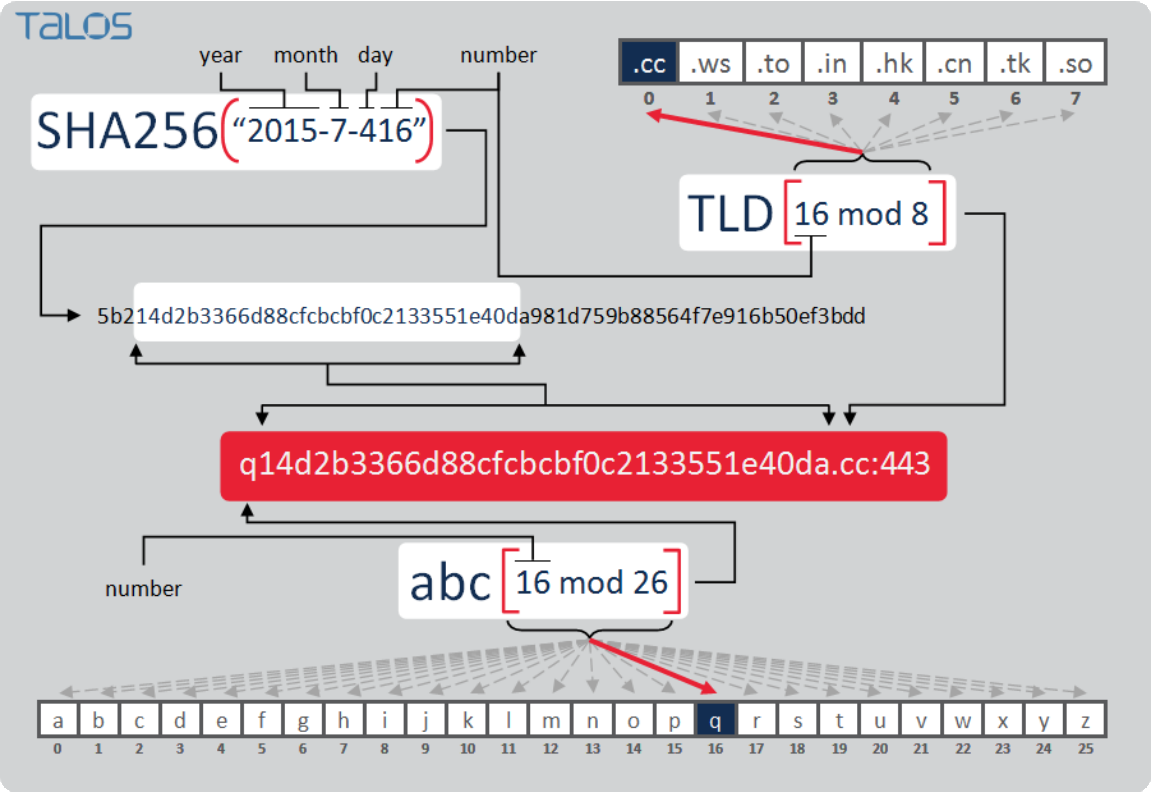
Why DNS is so valuable – DGA Detection

- Frequently used by malware for dynamic C&C generation
- Example
  - Dyre DGA algorithm
  - Provided by Talos



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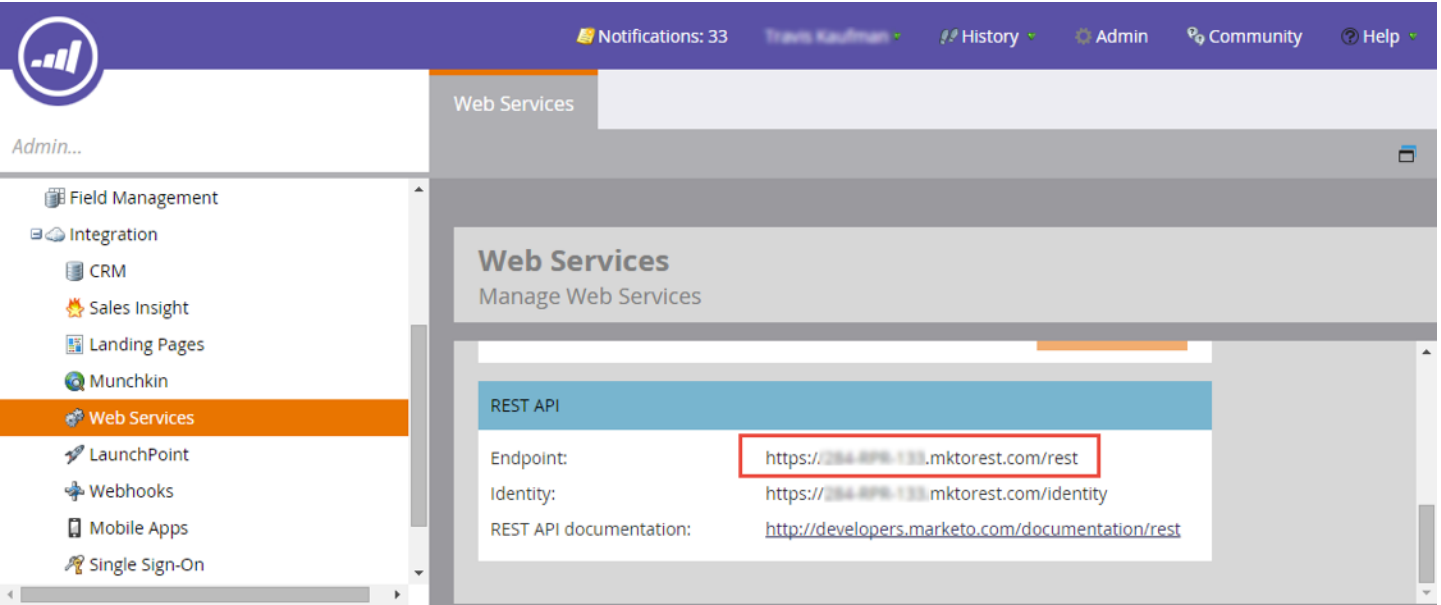
Why DNS is so valuable – DGA Detection

- Also used for marketing campaigns
  - e.g. Marketo
  - http://**100-AEK-913.mktorest.com**/rest/v1/lead/{id}.json



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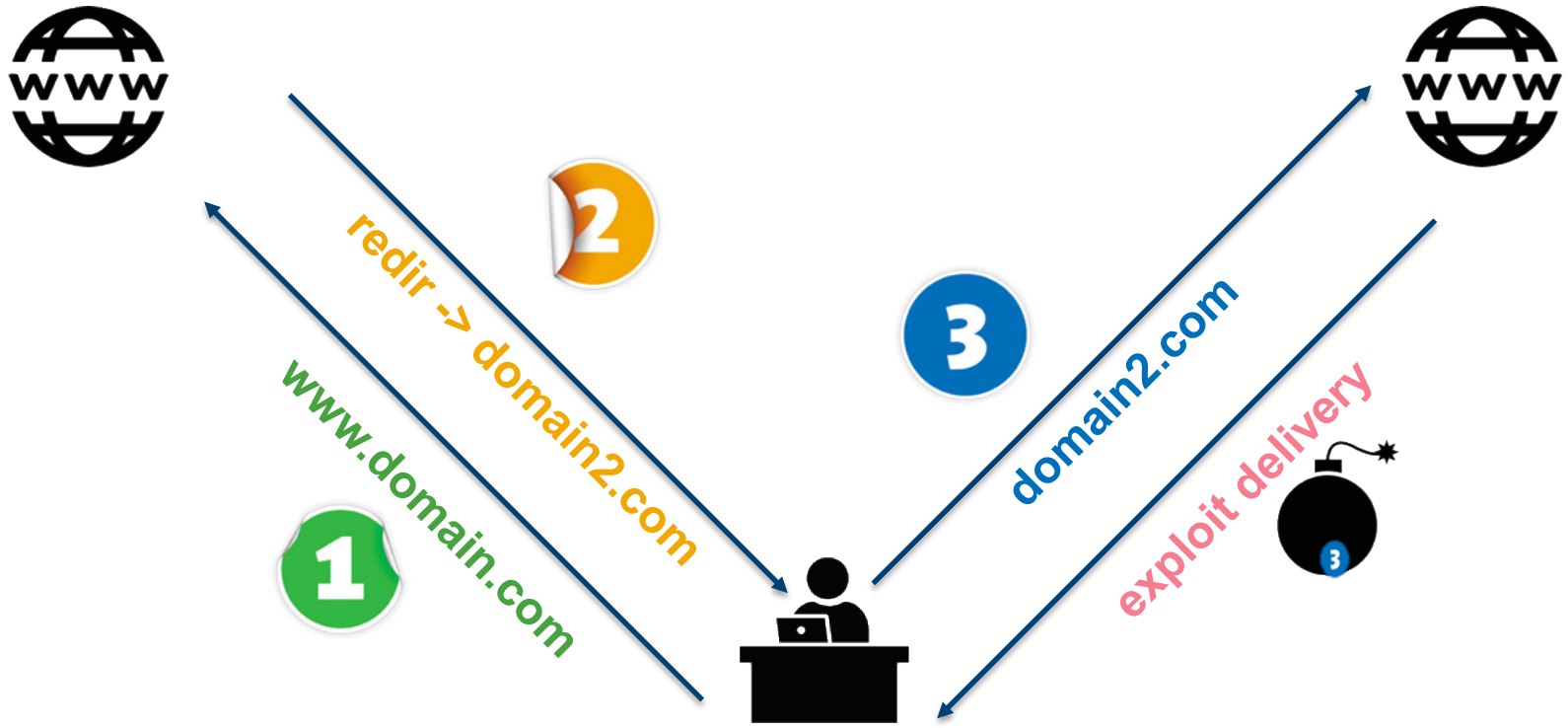
Why DNS is so valuable – Redirects



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Real world example

Of using DNS for threat hunting and incident response

Real World Example: Xerox Printers

- Noticed a rather odd domain name being beaconed out to: xeroxdiscoverysupernode3.com
  - Malware?
  - Phishing?
  - Drive update site?
  - Too long to be valid, right?
- Domain was not even registered!
  - Used OpenDNS Investigate
  - xeroxdiscoverysupernode[1-3].com were each seeing ~2,000 queries per hour

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Real World Example: Xerox Printers

- Google showed these domains as being related to Xerox® WorkCentre® 5845/5855/5865/5875/5890 and 7000-series line of printers
- Origin of beaconing was from the McAfee ePolicy Orchestrator (ePO) security module

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Real World Example: Xerox Printers

WorkCentre™ 5865/5875/5890

New!
Xerox ConnectKey Controller
Copy, print, scan, fax, email
Speed: up to 65 / 75 / 90 ppm
Maximum paper capacity: 8,700 sheets

Learn more

Compare

List Price
$25,245

Purchase Options +

WorkCentre™ 7755/7765/7775

Copy, print with optional scan, fax, email
Best for high-volume jobs of any size with great color when you need it
Color: up to 40 / 50 ppm
Black: up to 55 / 65 / 75 ppm
Standard paper capacity: 3,260 sheets
Maximum paper capacity: 5,260 sheets

Learn more

Compare

List Price
$34,100

Purchase Options +

Xerox® Color 560/570

Copy, print, scan, fax, email
Best for graphics-intensive and short-run production applications
Color: up to 60/70 ppm
Black: up to 65/75 ppm
Maximum paper capacity: 7,260 sheets

Learn more

Compare

List Price
$39,995

Purchase Options +

Not inexpensive printers…

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Real World Example: Xerox Printers

- So what were they doing?

- They were trying to do a handshake
  - …over the Internet
  - …because that's where DNS said the server was.

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Real World Example: Xerox Printers

- Printers querying from all over the world!

- Long story short…
  - Xerox printers querying out to the Internet
  - Documentation says to configure the xeroxdiscoverysupernode[1-3] domains to point internally
  - Obviously not everyone read the instructions

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Real World Example: Xerox Printers

- Registered the domains, pointed to a VPS, and monitored traffic.

Bytes

Bytes	Time	Value
Min	04/21/14 10:00:00	0 bps
Max	04/21/14 13:00:00	1.25 Mbit
Last	04/22/14 19:37:46	43.8 bps
Average	474.76 Kbit	
Total Traffic	2.65 GB	
Selection Time	Mon Apr 21 2014 09:00:00 GMT-0700 (PDT)	
Minute Top Talkers	• Senders [Avg Traffic/sec] 1. [redacted] (22 Kbit) 2. [redacted] (22 Kbit) 3. [redacted] (5 Kbit) • Receivers [Avg Traffic/sec] 1. [redacted] (4 Kbit) 2. [redacted] (4 Kbit) 3. [redacted] (4 Kbit)	

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Real World Example: Xerox Printers

- Domains sinkholed (at OpenDNS)
- Blog post released
  - http://labs.opendns.com/2014/05/01/xerox-printer-beacons/
- All fixed now, right?

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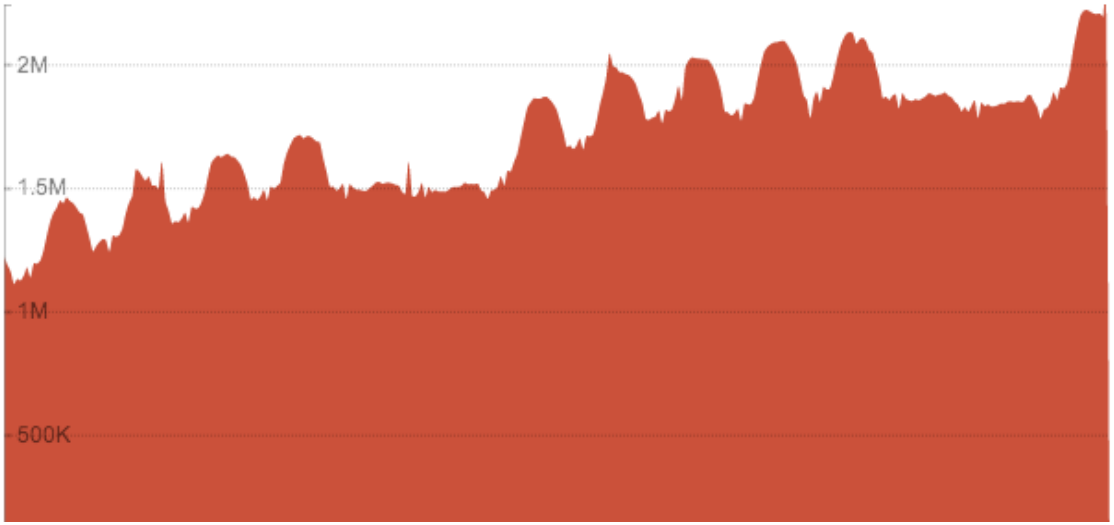
Real World Example: Xerox Printers

NOTE: Click on the graph to zoom.



Bytes

Bytes	Time	Value
Min	05/05/14 19:00:00	0 bps
Max	05/05/14 12:00:00	2.12 Mbit
Last	05/05/14 19:03:51	75.4 bps
Average	1.62 Mbit	
Total Traffic	240.09 GB	
Selection Time	Wed Apr 30 2014 17:00:00 GMT-0700 (PDT)	

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Available tools

Free and commercial

Free Tools

- PassiveTotal
  - Passive DNS threat-analysis platform
  - https://www.passivetotal.org/
- Site24x7
  - Generate a DNS Report for a domain
  - https://www.site24x7.com/dns-lookup.html
- DNSstuff Toolbox
  - Various tools related to DNS intelligence
  - http://www.dnsstuff.com/tools

Free Tools (Continued…)

- Domaintools
  - Whois lookup, other features for a price
  - http://research.domaintools.com/
- Elasticsearch, Logstash, Kibana (ELK)
  - Take data from any source, any format and search, analyze, and visualize it
  - https://www.elastic.co/
- Graylog
  - "Open source log management that actually works."
  - https://www.graylog.org/

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Commercial Tools

- PassiveTotal Enterprise (by RiskIQ)
  - Geared towards the enterprise
  - https://www.passivetotal.org/enterprise
- Domaintools
  - Domain, DNS and Internet OSINT-based cyber threat intelligence (14yrs worth)
  - http://domaintools.com/
- OpenDNS Investigate
  - View of Internet domains, IPs, and ASNs
  - https://www.opendns.com/enterprise-security/threat-intelligence/

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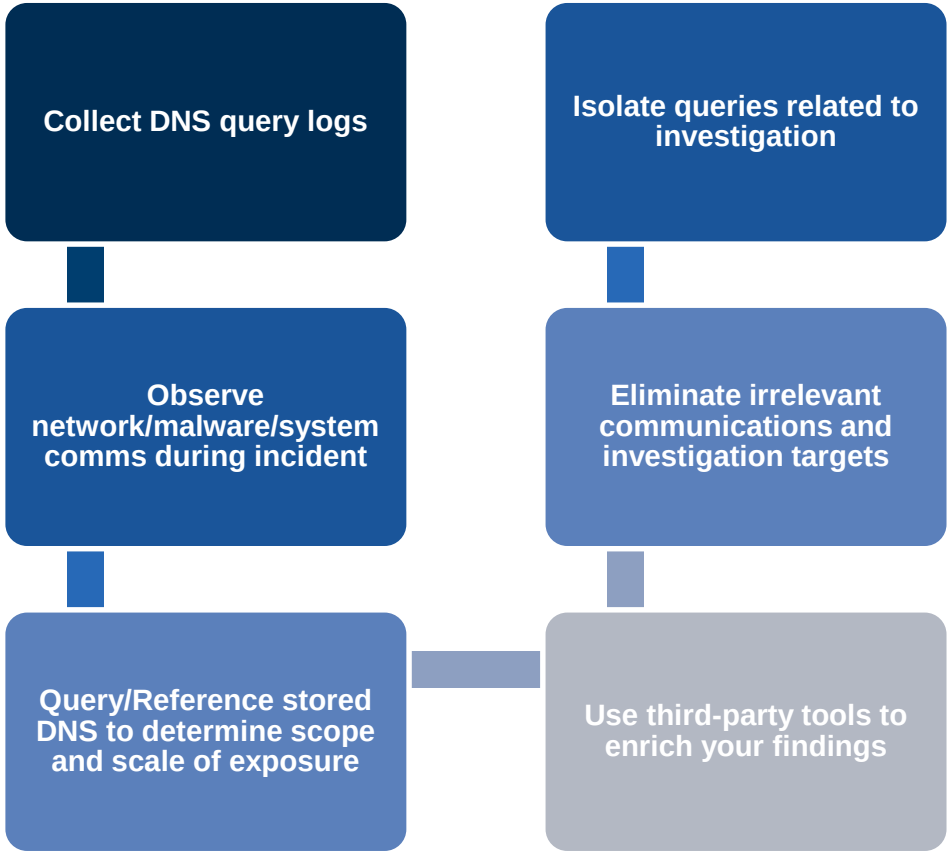
Summary

This is the end my friends...

But Before We Go…



- I wanted to leave you with a workflow for THIR w/DNS
- Hopefully this helps you wrap your heads around how to employ DNS logs during investigations

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Summary

- DNS is one of the core protocols used on the Internet
  - And by Internet-connected devices
  - So…like…use it to your benefit, bro!
- DNS can be used to focus your investigation and create a timeline of related events/actions
- Wealth of tools (free and commercial) available to help generate intelligence and corroborate information for use in your investigations

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Questions?

Contact me at
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